PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The Impact of Levothyroxine in Women with Positive Thyroid Antibodies on Pregnancy Outcomes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials
AUTHORS	Lau, Lorraine; Benham, Jamie; Lemieux, Patricia; Yamamoto, Jennifer; Donovan, Lois

VERSION 1 – REVIEW

REVIEWER	Lifeng Lin Florida State University, USA
REVIEW RETURNED	31-Aug-2020

GENERAL COMMENTS	This manuscript presents a systematic review and meta-analysis on the effect of levothyroxine therapy on pregnancy outcomes in women without overt hypothyroidism who are TPOAb and/or TgAb-positive. I focused on reviewing the statistical analyses, and my comments are as follows. Page 9, line 10. Change “Random-effect” to “Random-effects.” “Relative risk and … were calculated.” This information was repeated, as the authors have stated that “meta-analyses were conducted to evaluate the relative risk of …” at the top of this page. Page 9, line 12. Change “I^2 statistics” to “I^2 statistic.” Please also provide citations for its interpretation (e.g., https://doi.org/10.1136/bmj.327.7414.557). Page 9, line 24. Specify that the graphical aids are the funnel plot. Figure 1. The results in the steps from “Records screened” to “Full-text articles assessed for eligibility” do not match. The authors screened 2812 records and excluded 1393 records; there should be 1419 records for full-text assessment, not just 79 records. Also, in the rightmost box in the step of “Eligibility,” indicate that “Not a randomized controlled trials” and the following two items belonged to the “Full-text articles excluded.” In the forest plots (Figures 3 and Figures S4 and S5), “risk ratio” was used, but “relative risk” was used in the main text. Please be consistent. Page 15, line 19. “On visual inspection, there was asymmetry indicating potential publication bias.” Please indicate the direction of potential bias. Please also be aware of the limitations of the funnel plot when the number of studies is small (as in the meta-analyses presented in this manuscript); see https://doi.org/10.1136/bmj.d4002. In the funnel plot (Figure S6),

	please change the x- and y-labels to be informative phrases, rather than the variable name used in coding.

REVIEWER	René Rodriguez Gutierrez Mayo Clinic and UANL, USA and Mexico
REVIEW RETURNED	29-Sep-2020

GENERAL COMMENTS	Lau et al. carried out a systematic review and metanalysis of randomized controlled trials with the objective to evaluate the effect of levothyroxine therapy in women without overt hypothyroidism with positive thyroid/thyroglobulin antibodies on pregnancy outcomes. This is a relevant question since the ATA in 2017 added to their guidelines the recommendation on levothyroxine on these patients. The authors found no significant difference in the risk of the included pregnancy outcomes between the levothyroxine group compared to the control group. Several previous systematic reviews, which included both observational and experimental studies, have had inconsistent results. This current study aims to reduce the heterogeneity in the study design by including only randomized clinical trials. After a critical evaluation based on the AMSTAR assessment tool, the systematic review is judged at high quality with one critical flaw which is failure to discuss and consider the risk of bias of the included studies in the interpretation and conclusions of their results. Additionally, there are a few other suggestions that could further augment the quality of the manuscript, they are proposed as follows: 1. Please specify whether the authors searched the reference list of the included articles to further identify potentially relevant studies. 2. If available, the authors should consider including as an appendix a list of the excluded studies justifying their exclusion. 3. In page 5, line 43, the authors should support the statement of "...improves pregnancy outcomes in women with thyroid autoimmunity is controversial" with the appropriate citations despite the fact that further on in the discussion section the authors provide the relevant citations. 4. In Table 1 it would be valuable if the authors add the duration and dose treatment of the therapy in each study. Also, it would be helpful if the authors could be more specific in describing patient's demographic characteristics and source of funding in each study. As a sensitivity analysis, authors could further explore if estimates vary when dividing studies by the source of funding. 5. On page 7 line 24 the authors go into the details of the study selection stages. Before each stage, was a pilot carried out to calibrate researcher's criteria? 6. In the pregnancy outcomes section, the PROSPERO registration includes an additional outcome of childhood neurodevelopment. Was this outcome not possible to evaluate? Including additional details about this adjustment would increase transparency and instill confidence in the reader. 7. In Figure 1, the records excluded, and full-text articles assessed do not add up as there are about 1400 articles unaccounted for in the diagram. The authors are encouraged to edit this figure to reflect the process more accurately. 8. As a minor revision, regarding Supplemental Table (S2), adding a legend to provide information about each blank space to specify if data was missing or not available could make this table simpler to peruse.

	9. As a minor revision, regarding Supplemental Figure (S6), the x and y labels appear to be unedited. 10. As a minor revision, on page 17 line 6, it appears that the format for citation 27 is perhaps not consistent or perhaps not necessary in this sentence. 11. The authors mention that they used a modified version of the Cochrane Risk of Bias Tool. It would be appropriate to describe the characteristics and rationale behind the modification. 12. Referring to the last paragraph in the discussion section, randomized controlled trials do not always provide strong evidence to guide clinical decisions. In order for this to be the case, they should have good methodological quality and the body of evidence should be consistent, precise, and the clinical trials should evaluate relevant outcomes. The GRADE approach provides a methodology to assess the overall confidence in the body of evidence. The authors should consider adopting this approach to summarize the quality of their included studies and the confidence they can provide for each of the evaluated outcomes. The results from the GRADE assessment should be further incorporated in the conclusion and abstract of the study.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Lifeng Lin
Institution and Country: Florida State University, USA
Please state any competing interests or state 'None declared': None declared.

This manuscript presents a systematic review and meta-analysis on the effect of levothyroxine therapy on pregnancy outcomes in women without overt hypothyroidism who are TPOAb and/or TgAb-positive. I focused on reviewing the statistical analyses, and my comments are as follows.

Thank you for Dr. Lin's thorough and expert review of the statistical analyses.

Page 9, line 10. Change "Random-effect" to "Random-effects." "Relative risk and … were calculated." This information was repeated, as the authors have stated that "meta-analyses were conducted to evaluate the relative risk of …" at the top of this page.

Thank you for this helpful advice. Yes, this is now addressed on page 9, lines 24-26, and line 31.

Page 9, line 12. Change "I^2 statistics" to "I^2 statistic." Please also provide citations for its interpretation (e.g., https://doi.org/10.1136/bmj.327.7414.557).

Thank you, addressed on page 9, line 33.

Page 9, line 24. Specify that the graphical aids are the funnel plot.

Thank you, addressed on page 9, line 45.

Figure 1. The results in the steps from "Records screened" to "Full-text articles assessed for eligibility" do not match. The authors screened 2812 records and excluded 1393 records; there should be 1419 records for full-text assessment, not just 79 records. Also, in the rightmost box in the step of "Eligibility," indicate that "Not a randomized controlled trials" and the following two items belonged to the "Full-text articles excluded."

Thank you, corrected in Figure 1.

In the forest plots (Figures 3 and Figures S4 and S5), "risk ratio" was used, but "relative risk" was used in the main text. Please be consistent.

Good point, thank you for this correction. This has been corrected in Figures 3, S4 and S5.

Page 15, line 19. "On visual inspection, there was asymmetry indicating potential publication bias." Please indicate the direction of potential bias. Please also be aware of the limitations of the funnel plot when the number of studies is small (as in the meta-analyses presented in this manuscript); see https://doi.org/10.1136/bmj.d4002. In the funnel plot (Figure S6), please change the x- and y-labels to be informative phrases, rather than the variable name used in coding.

Thank you, corrected on page 16, lines 3-5 to the following: "On visual inspection, there was asymmetry indicating potential publication bias towards smaller studies reporting positive findings."

The following limitations were added to page 19, lines 13-17: "Lastly, the funnel plot suggests publication bias towards smaller studies reporting positive findings. Interpretation of the funnel plot is limited by the small number of studies included in this meta-analysis[35]."

Figure S6 has been corrected to the correct variable names.

Reviewer: 2
Reviewer Name: René Rodriguez Gutierrez
Institution and Country: Mayo Clinic and UANL, USA and Mexico
Please state any competing interests or state 'None declared': None

Lau et al. carried out a systematic review and metanalysis of randomized controlled trials with the objective to evaluate the effect of levothyroxine therapy in women without overt hypothyroidism with positive thyroid/thyroglobulin antibodies on pregnancy outcomes. This is a relevant question since the ATA in 2017 added to their guidelines the recommendation on levothyroxine on these patients. The authors found no significant difference in the risk of the included pregnancy outcomes between the levothyroxine group compared to the control group. Several previous systematic reviews, which included both observational and experimental studies, have had inconsistent results. This current study aims to reduce the heterogeneity in the study design by including only randomized clinical trials. After a critical evaluation based on the AMSTAR assessment tool, the systematic review is judged at high quality with one critical flaw which is failure to discuss and consider the risk of bias of the included studies in the interpretation and conclusions of their results. Additionally, there are a few

other suggestions that could further augment the quality of the manuscript, they are proposed as follows:

Thank you to Dr. Gutierrez for his comprehensive review and comments. In addressing the critical flaw that was highlighted, we have addressed the risk of bias (as determined by Cochrane Risk of Bias Tool) in the limitations section of our discussion. Furthermore, we employed the GRADE assessment as recommended to summarise the quality of evidence from this systematic review.

1. Please specify whether the authors searched the reference list of the included articles to further identify potentially relevant studies.

The following has been added to the manuscript on page 8, lines 3-6.

"The reference list of included articles was searched for the addition of further relevant studies."

2. If available, the authors should consider including as an appendix a list of the excluded studies justifying their exclusion.

Reasons for exclusion of full-text articles are summarised in Figure 1 as per PRISMA guidelines.

3. In page 5, line 43, the authors should support the statement of "...improves pregnancy outcomes in women with thyroid autoimmunity is controversial" with the appropriate citations despite the fact that further on in the discussion section the authors provide the relevant citations.

Thank you. These citations have been inserted on page 5, lines 46.

4. In Table 1 it would be valuable if the authors add the duration and dose treatment of the therapy in each study. Also, it would be helpful if the authors could be more specific in describing patient's demographic characteristics and source of funding in each study. As a sensitivity analysis, authors could further explore if estimates vary when dividing studies by the source of funding.

Duration of treatment, dosing and titration and source of funding are now provided in Supplemental Table S2. There was insufficient information to stratify the trials by source of funding.

5. On page 7 line 24 the authors go into the details of the study selection stages. Before each stage, was a pilot carried out to calibrate researcher's criteria?

Yes, prior to each selection stage, a calibration step was included. This following was added to the methods section on page 8, lines 3-6.

"Prior to each selection step, a pilot calibration was conducted by the respective reviewers to review inclusion criteria."

6. In the pregnancy outcomes section, the PROSPERO registration includes an additional outcome of childhood neurodevelopment. Was this outcome not possible to evaluate? Including additional details about this adjustment would increase transparency and instill confidence in the reader.

Thank you for this. The following has been added to the manuscript on page 11, lines 3-6

"There was insufficient information to assess the outcome of childhood neurodevelopment."

7. In Figure 1, the records excluded, and full-text articles assessed do not add up as there are about 1400 articles unaccounted for in the diagram. The authors are encouraged to edit this figure to reflect the process more accurately.

Thank you. This has been corrected in Figure 1.

8. As a minor revision, regarding Supplemental Table (S2), adding a legend to provide information about each blank space to specify if data was missing or not available could make this table simpler to peruse.

Thank you. This has been addressed in Supplemental Table S2.

9. As a minor revision, regarding Supplemental Figure (S6), the x and y labels appear to be unedited.

Thank you. This has been addressed on Supplemental Figure S6.

10. As a minor revision, on page 17 line 6, it appears that the format for citation 27 is perhaps not consistent or perhaps not necessary in this sentence.

Thank you. This has been corrected on page 17, line 45.

11. The authors mention that they used a modified version of the Cochrane Risk of Bias Tool. It would be appropriate to describe the characteristics and rationale behind the modification.

The characteristics and rationale for modification of the Cochrane Risk of Bias Tool are summarised in Higgins JPT GS. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. Available from http://handbook.cochrane.org.

The citation has been added to this manuscript on page 8, line 45. We did not make any modifications to the published modified Cochrane Risk of Bias Tool.

12. Referring to the last paragraph in the discussion section, randomized controlled trials do not always provide strong evidence to guide clinical decisions. In order for this to be the case, they should have good methodological quality and the body of evidence should be consistent, precise, and the clinical trials should evaluate relevant outcomes. The GRADE approach provides a methodology to assess the overall confidence in the body of evidence. The authors should consider adopting this approach to summarize the quality of their included studies and the confidence they can provide for each of the evaluated outcomes. The results from the GRADE assessment should be further incorporated in the conclusion and abstract of the study.

This is a good point. Two authors (LD and LL) reviewed the GRADE assessments for the four main study outcomes. The results are now incorporated in the abstract, methods, results, and discussion sections.

Thank you again for taking the time to review our manuscript. Should you have any questions or require further information, I may be reached by phone at (403) 955-8358 or by e-mail at lois.donovan@albertahealthservices.ca

VERSION 2 – REVIEW

REVIEWER	René Rodriguez Gutierrez UANL and Mayo Clinic
REVIEW RETURNED	11-Dec-2020

GENERAL COMMENTS	Lau et al performed a systematic review and meta-analysis of randomized clinical trials assessing the effect of levothyroxine therapy on pregnancy outcomes in patients with positive thyroid antibodies but without overt hypothyroidism, compared with placebo. The research question aligns with the scope of the journal, and the conclusions derived from the study are important for patients, clinicians, and policymakers, as the results fail to

	demonstrate a significant benefit on pregnancy outcomes among women treated with levothyroxine. The critical appraisal of this study was performed with AMSTAR-2 tool which renders for a high overall confidence in the results. The manuscript is well-written and adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidance, nonetheless there are some details that could further improve the quality of the manuscript: 1. As a minor observation, some abbreviations are not clarified in the abstract section. It would be beneficial to describe them to make it clearer for the readers. 2. In page 7, line 34, the authors state April,17, 2019 as the final date for study selection instead of November,5, 2020. 3. In page 8, line 44, the authors mention that they used a modified version of the Cochrane Risk of Bias Tool, but cite the Cochrane Handbook. It would be appropriate to add the reference of the modified version, and to describe the rationale behind this decision. 4. If applicable, could the authors please describe how was missing data handled in the Data Collection or Data Synthesis sections? 5. Could the authors please specify if the TSH levels presented on Table 1 are mean or median? 6. As a recommendation, the authors should consider improving the quality of the figures presented since they provide very important information for the readers. 7. There is a typo on Figure 1 when referencing the excluded articles, the figure read “Full-test articles” instead of full-text. 8. In Supplemental Table (S2), the authors should consider to report “Duration of treatment” with more detail since the difference between “until the end of pregnancy” and “throughout pregnancy” is not clear. 9. A considerable strength in this article is the use of GRADE to evaluate the quality of the evidence, which is why it would be very valuable if the authors could include GRADE’s summary of findings table in the manuscript.

VERSION 2 – AUTHOR RESPONSE

Thank you to the editor and reviewers for their helpful comments that have improved this manuscript. The comments are listed below in bold followed by our responses.

1. As a minor observation, some abbreviations are not clarified in the abstract section. It would be beneficial to describe them to make it clearer for the readers.

Thank you. These changes to the abstract have been made.

2. In page 7, line 34, the authors state April,17, 2019 as the final date for study selection instead of November,5, 2020.

Thank you. This update has been made.

3. In page 8, line 44, the authors mention that they used a modified version of the Cochrane Risk of Bias Tool, but cite the Cochrane Handbook. It would be appropriate to add the reference of the modified version, and to describe the rationale behind this decision.

Thank you. The word “modified” has been removed from the manuscript, as we used the original tool. A modified tool was developed in 2020, which we did not use as it is just being rolled out. The correct reference below has been added.
Higgins Julian P T, Altman Douglas G, Gøtzsche Peter C, Jüni Peter, Moher David, Oxman Andrew Det al. The Cochrane Collaboration’s tool for assessing risk of bias in randomised trials BMJ 2011; 343 :d5928

4. If applicable, could the authors please describe how was missing data handled in the Data Collection or Data Synthesis sections?

To be included in the systematic review, trials had to meet all inclusion criteria including reporting on pregnancy outcomes. Trial level data was collected for all pre-specified outcomes from each trial in which it was reported. If a trial did not report on a pre-specified outcome, the trial was not included in the analysis for that outcome.

5. Could the authors please specify if the TSH levels presented on Table 1 are mean or median?

The TSH levels presented in Table 1 are means or median as provided in the original articles. This information is now provided in Table 1.

6. As a recommendation, the authors should consider improving the quality of the figures presented since they provide very important information for the readers.

The quality of the figures has been improved.

7. There is a typo on Figure 1 when referencing the excluded articles; the figure read “Full-test articles” instead of full-text.
Thank you. This typo has been fixed.

8. In Supplemental Table (S2), the authors should consider to report “Duration of treatment” with more detail since the difference between “until the end of pregnancy” and “throughout pregnancy” is not clear.

Table S2 has been edited to provide more specific information about the duration of treatment.

9. A considerable strength in this article is the use of GRADE to evaluate the quality of the evidence, which is why it would be very valuable if the authors could include GRADE’s summary of findings table in the manuscript.

The GRADE Table has been added to the manuscript.

VERSION 3 – REVIEW

REVIEWER	René Rodriguez-Gutierrez UANL and Mayo Clinic Mexico and USA
REVIEW RETURNED	06-Jan-2021

GENERAL COMMENTS	Thanks for making the suggested changes. No further comments.